Power BI Sales Analysis Dashboard

Introduction

This project focuses on analyzing sales performance using Microsoft Power BI. The objective is to transform raw sales data into meaningful insights by applying data transformation, calculated columns, DAX measures, and interactive visualizations. The dashboard helps monitor sales, profit, order count, target achievement, and regional performance, enabling better business decision-making.

Dataset

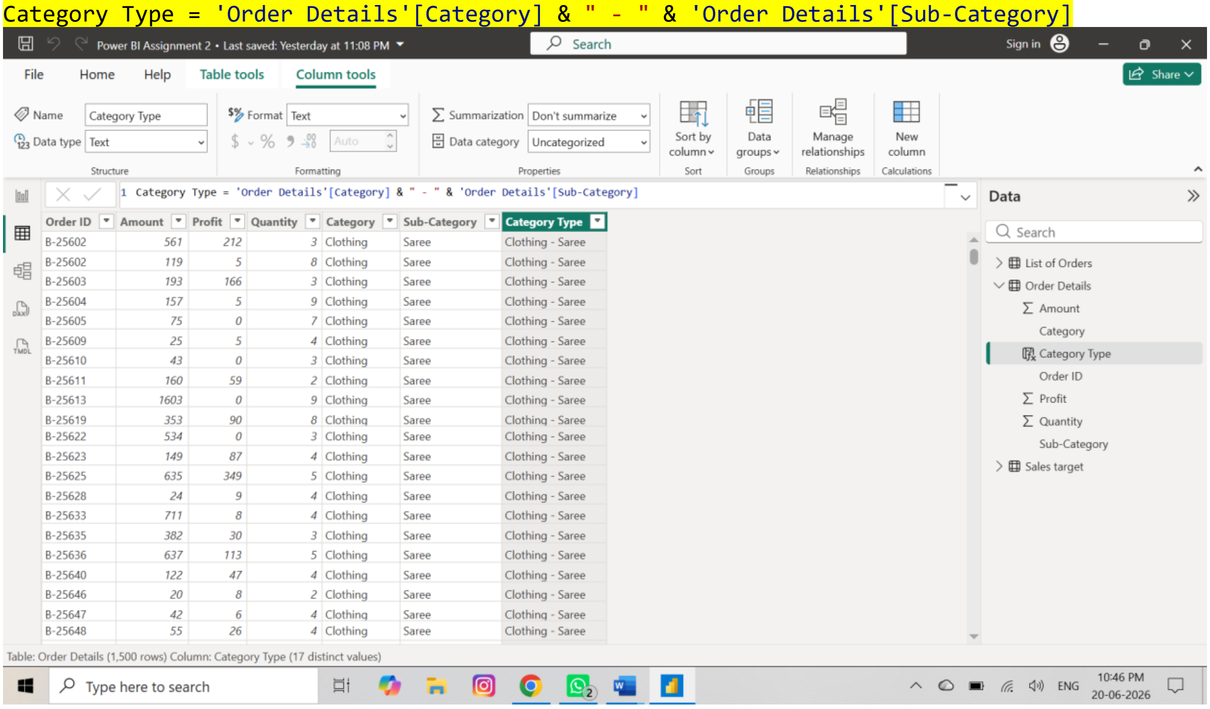
- List of orders
  Attributes: Order ID, Order Date, Customer Name, City, State, Category, Amount, Profit
- Order Details
  Attributes: Order ID, Category, Sub-Category, Quantity, Amount, Profit
- Sales Target
  Attributes: Category, Month of Order Date, Target

Data Analysis

DAX Calculations

- Create a Calculated Column for 'Category Type

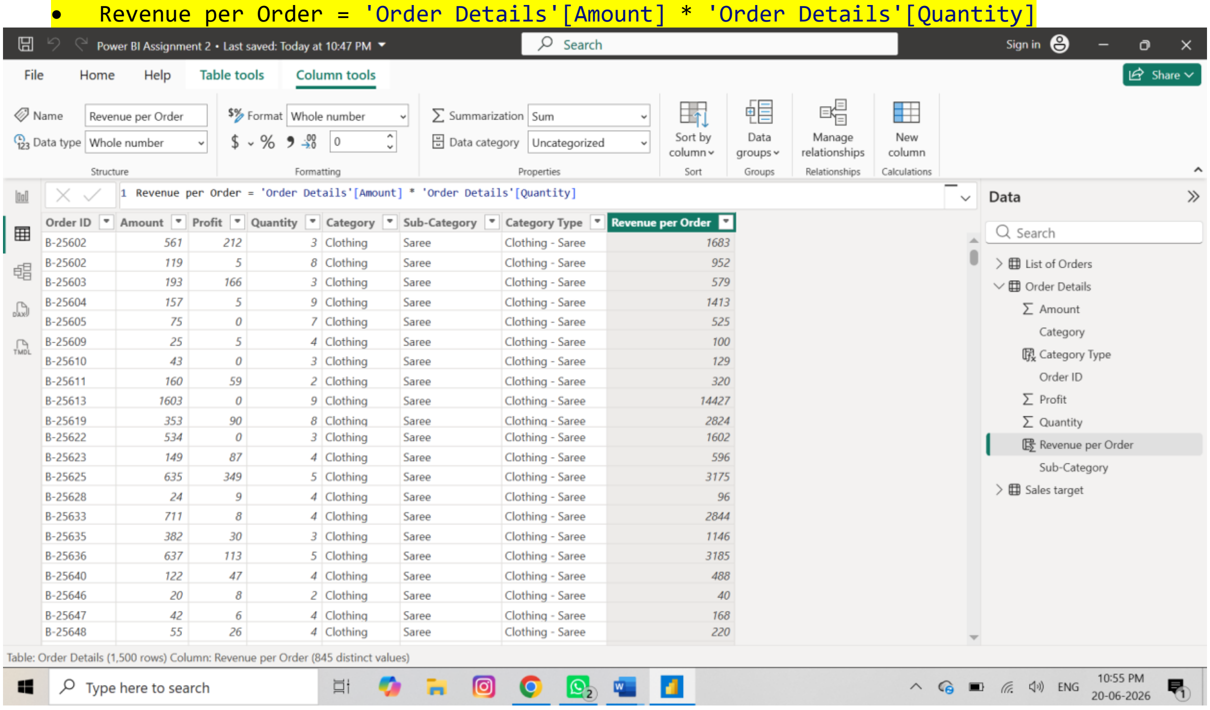
```
Category Type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]
```



Calculate Revenue per Order in Order Details Table

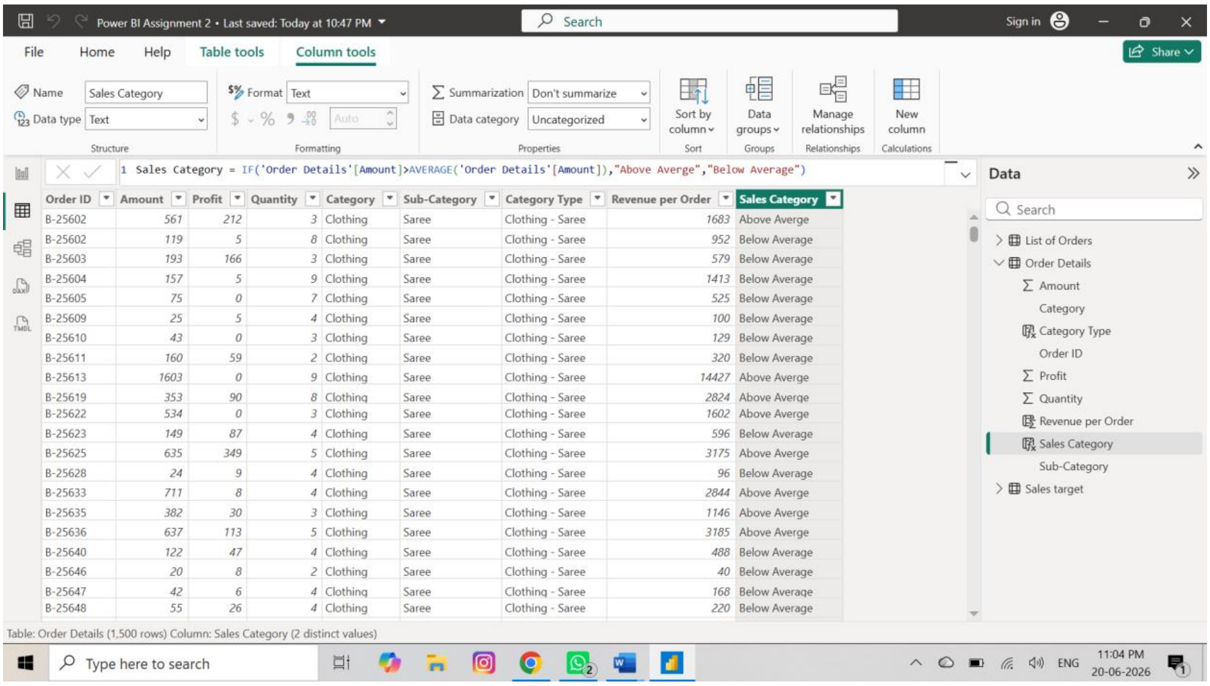
- Calculated Column – Revenue per Order

- `Revenue per Order = 'Order Details'[Amount] * 'Order Details'[Quantity]`



Create a Calculated Column to Categorize Sales

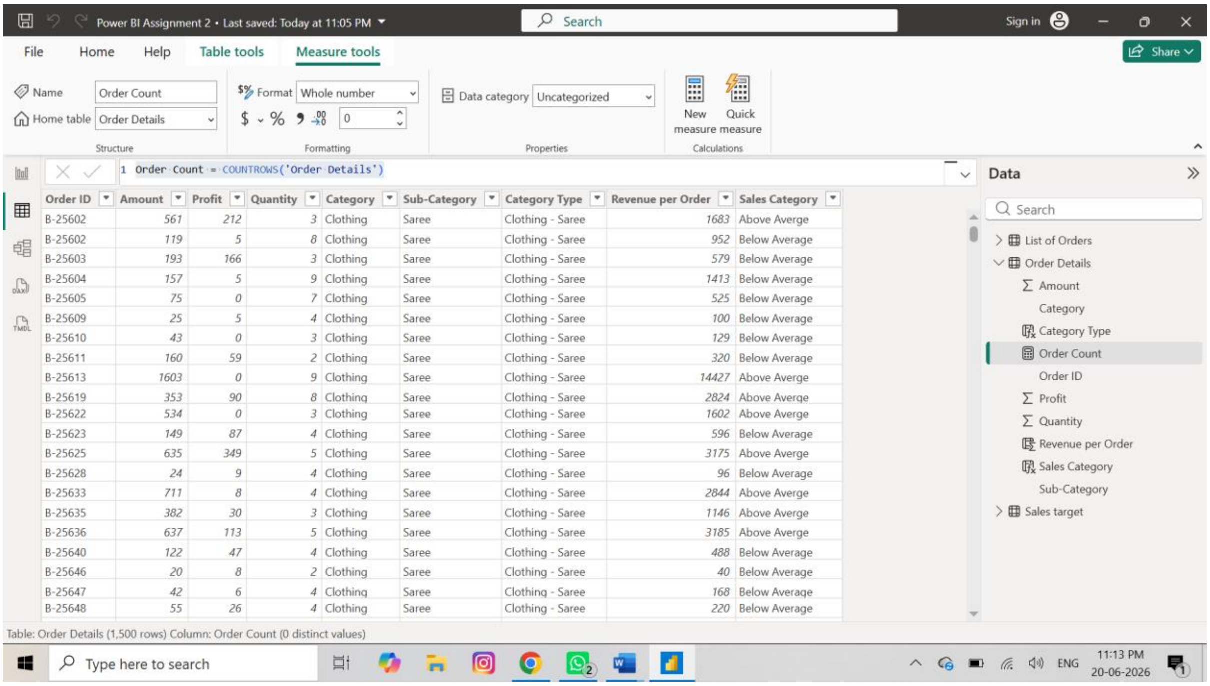
- **Calculated Column – Sales Category**
- `Sales Category = IF('Order Details'[Amount]>AVERAGE('Order Details'[Amount]),"Above Averge","Below Average")`



Calculated Measures

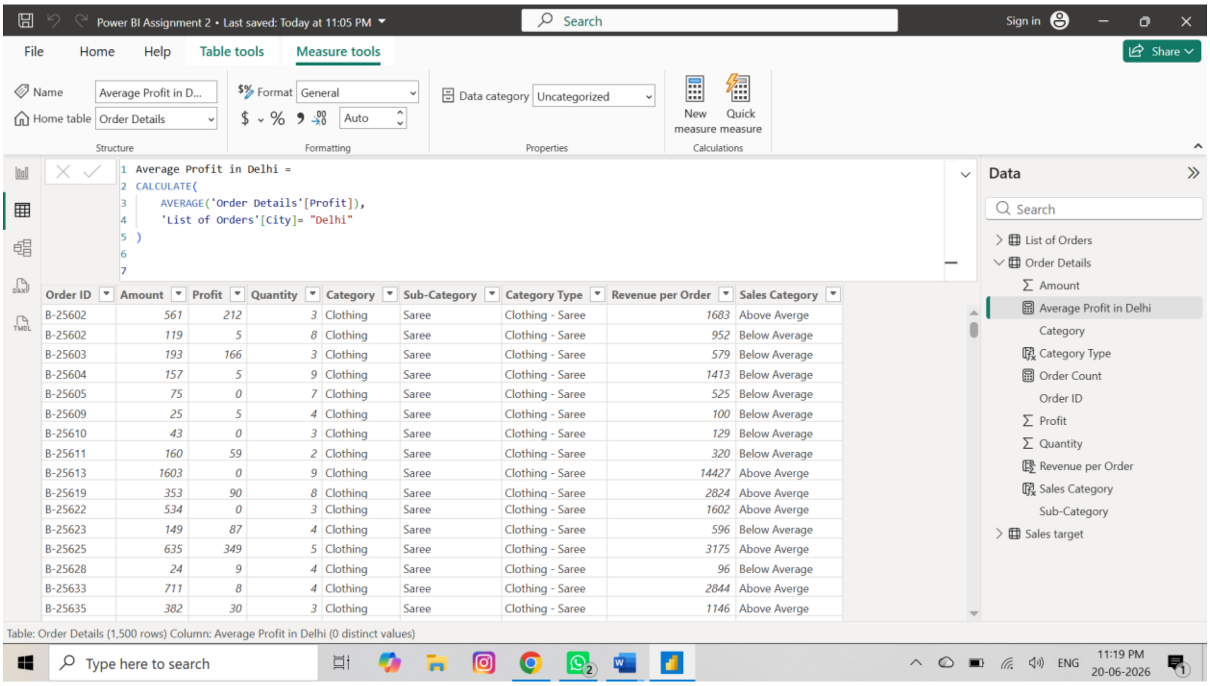
- **Order Count**

```
Order Count = COUNTROWS('Order Details')
```



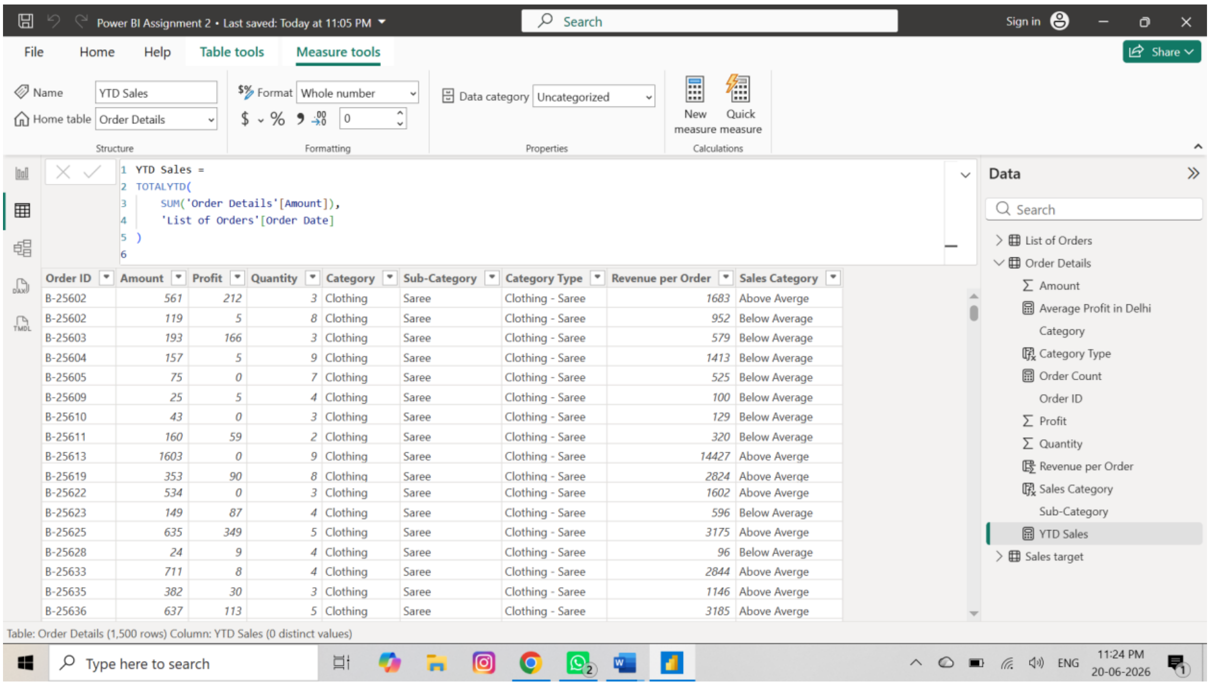
- **Average Profit in Delhi**

```
Average Profit in Delhi = CALCULATE( AVERAGE('Order Details'[Profit]), 'List of Orders'[City]= "Delhi")
```



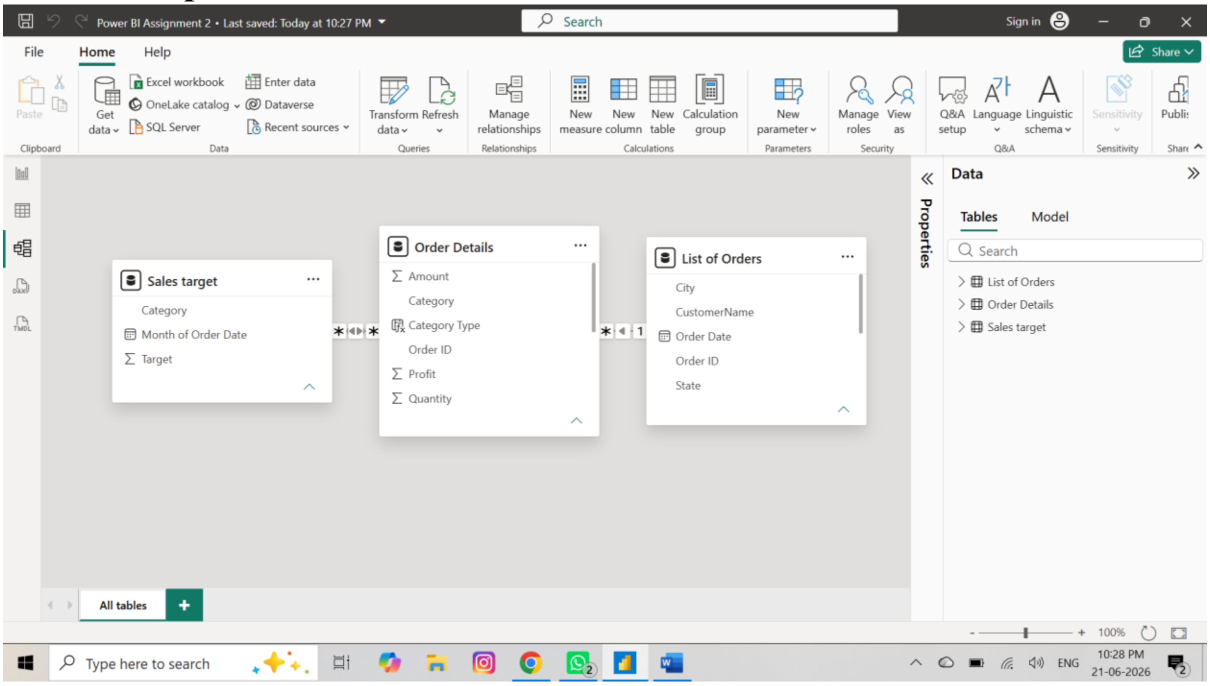
- **Year-to-Date (YTD) Sales**

```
YTD Sales = TOTALYTD( SUM('Order Details'[Amount]), 'List of Orders'[Order Date])
```



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	1683	Above Averge
B-25602	119	5	8	Clothing	Saree	Clothing - Saree	952	Below Average
B-25603	193	166	3	Clothing	Saree	Clothing - Saree	579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	1413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	525	Below Average
B-25609	25	5	4	Clothing	Saree	Clothing - Saree	100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	129	Below Average
B-25611	160	59	2	Clothing	Saree	Clothing - Saree	320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	14427	Above Averge
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	2824	Above Averge
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	1602	Above Averge
B-25623	149	87	4	Clothing	Saree	Clothing - Saree	596	Below Average
B-25625	635	349	5	Clothing	Saree	Clothing - Saree	3175	Above Averge
B-25628	24	9	4	Clothing	Saree	Clothing - Saree	96	Below Average
B-25633	711	8	4	Clothing	Saree	Clothing - Saree	2844	Above Averge
B-25635	382	30	3	Clothing	Saree	Clothing - Saree	1146	Above Averge
B-25636	637	113	5	Clothing	Saree	Clothing - Saree	3185	Above Averge

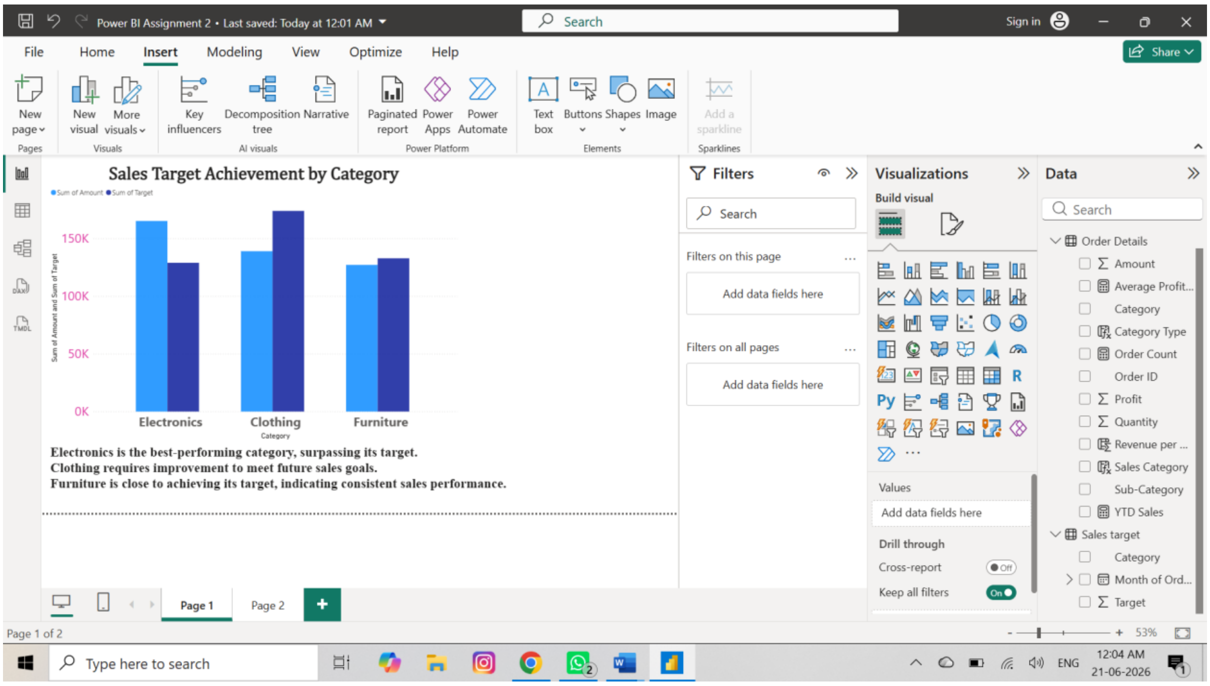
Relationship Connection



Data Visualizations

Sales Target Achievement by Category

Visual: Clustered Column Chart



My Analysis for Sales Target Achievement by Category

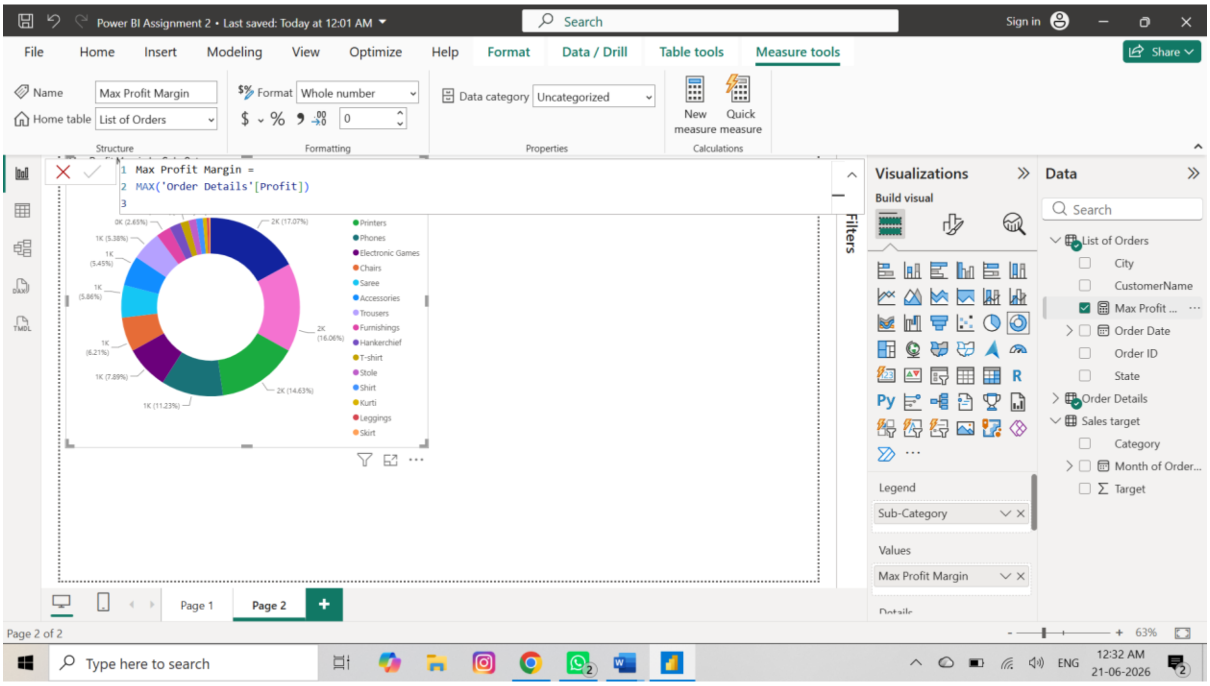
- Electronics is the best-performing category, surpassing its target.
- Clothing requires improvement to meet future sales goals.
- Furniture is close to achieving its target, indicating consistent sales performance.

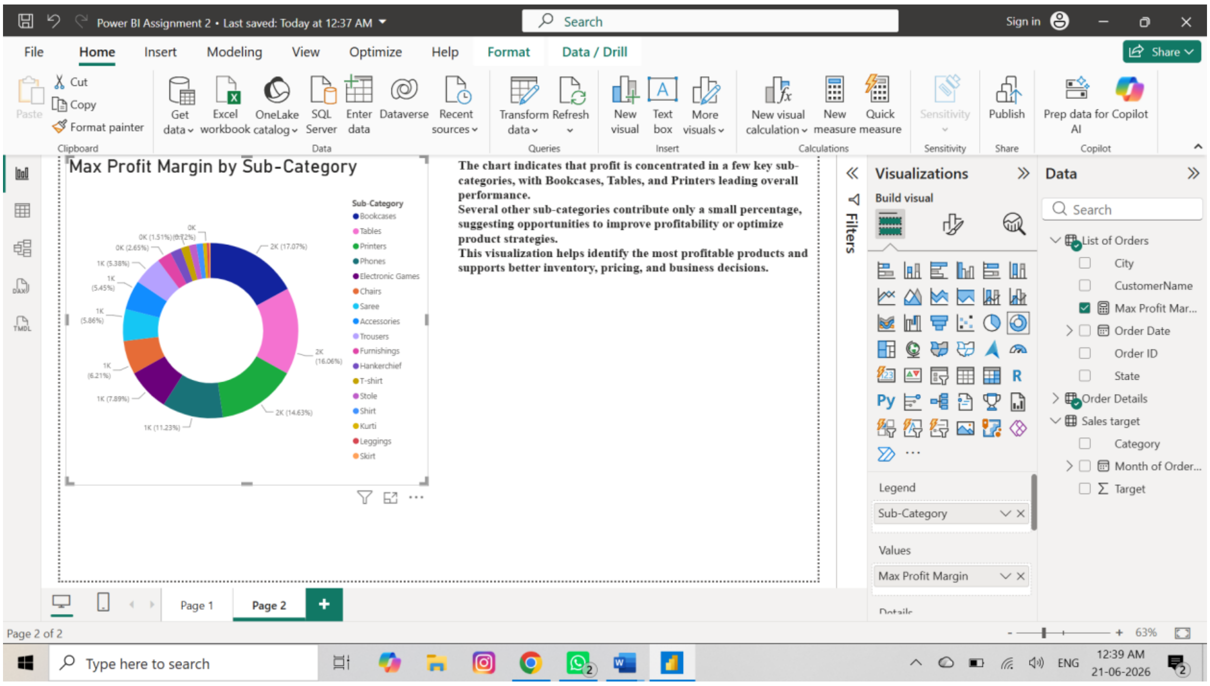
2. Max Profit Margin by Sub-Category

Create Measure:

```
Max Profit Margin =
MAX('Order Details'[Profit])
```

Visual: Donut Chart



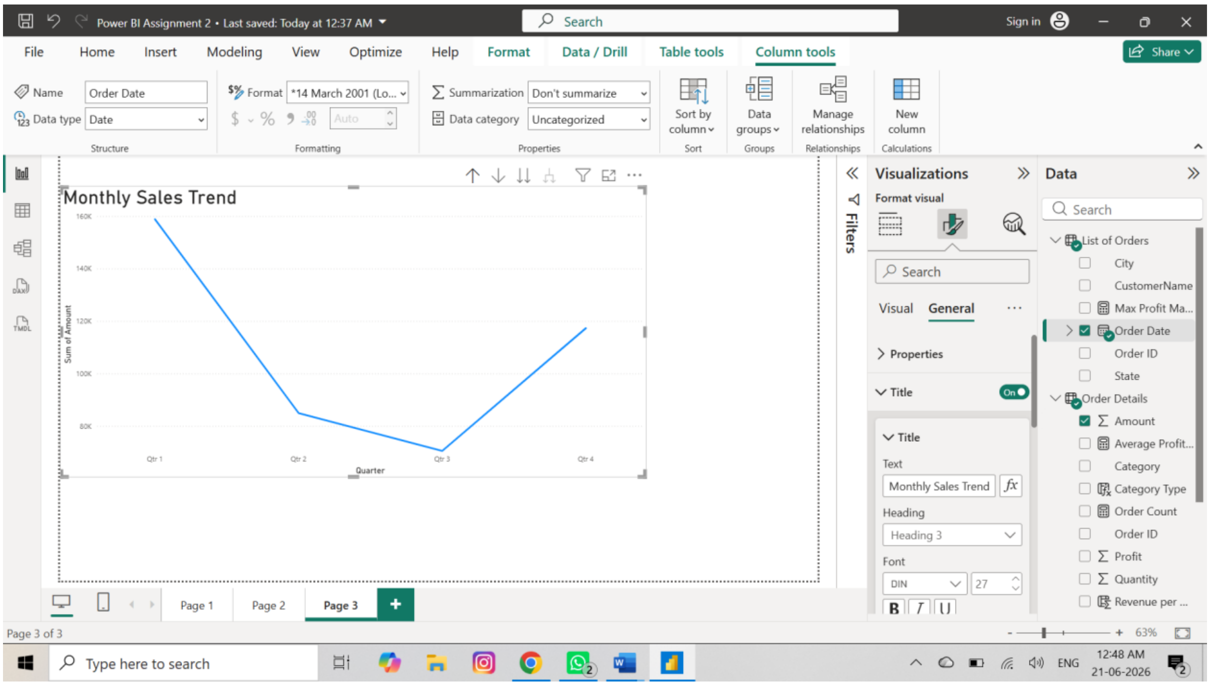


My Analysis Max Profit Margin by Sub-Category

- The chart indicates that profit is concentrated in a few key sub-categories, with Bookcases, Tables, and Printers leading overall performance.
- Several other sub-categories contribute only a small percentage, suggesting opportunities to improve profitability or optimize product strategies.
- This visualization helps identify the most profitable products and supports better inventory, pricing, and business decisions.

Monthly Sales Trend

Visual: Line Chart

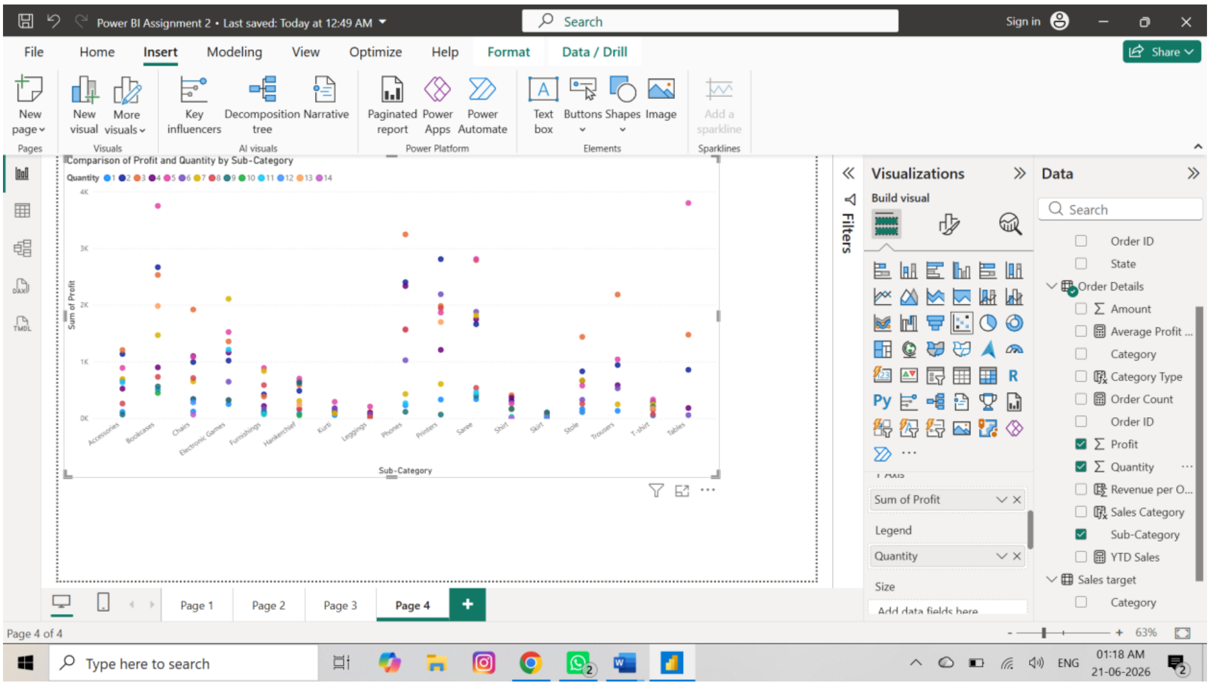


My Analysis For Monthly Sales Trend

- The business achieved its strongest performance in the first quarter.
- A downward trend is observed from Q1 to Q3, suggesting reduced demand or seasonal effects.
- The increase in Q4 indicates successful recovery, possibly due to promotional activities, festive sales, or increased customer demand.
- The chart helps identify seasonal patterns and supports planning for future sales strategies.

Comparison of Profit and Quantity by Sub-Category

- **Visual:** Scatter Chart



My Analysis for Comparison of Profit and Quantity by Sub-Category

- Bookcases, Printers, and Tables show the highest profit values, reaching above 3K.
- Chairs, Electronic Games, and Saree generate moderate profit levels across different quantities.
- Sub-categories such as Kurti, Leggings, Skirt, and Shirt contribute relatively lower profits.
- Profit values vary significantly within the same sub-category, indicating that quantity alone does not determine profitability.
- Several high-profit data points occur at specific quantities, suggesting that some products perform better at particular sales volumes.

Comparison of Total Sales Amount and Target

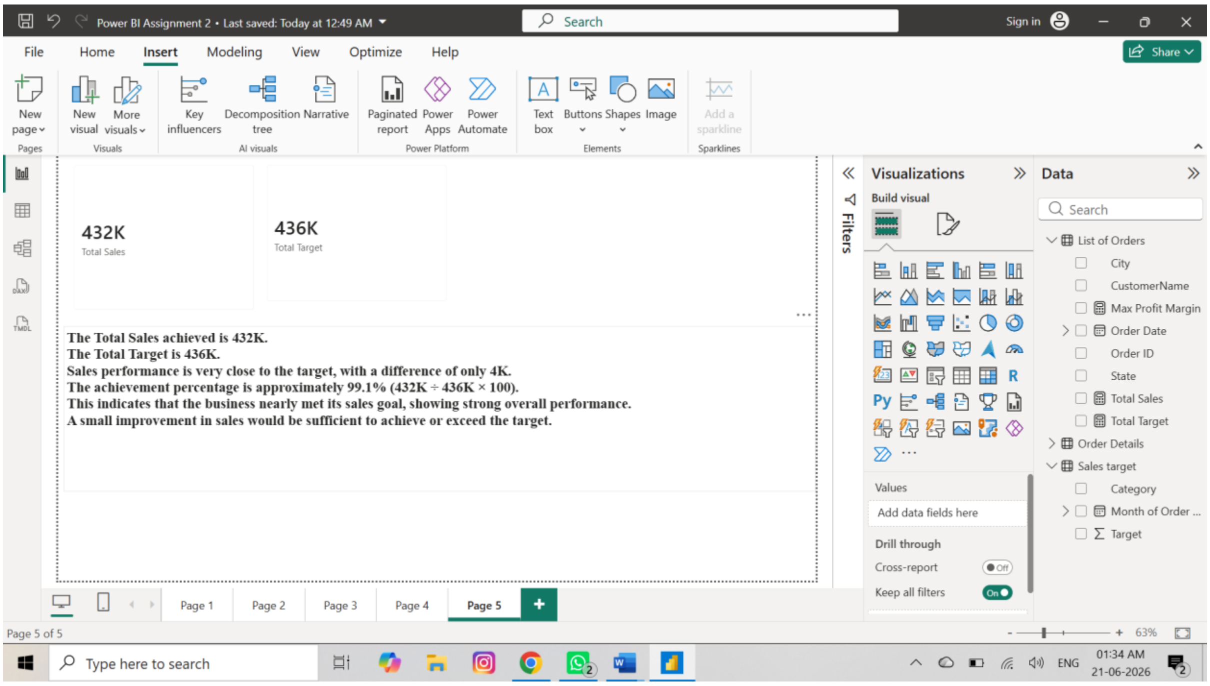
Measure 1

```
Total Sales =
SUM('Order Details'[Amount])
```

Measure 2

```
Total Target =
SUM('Sales Target'[Target])
```

Visual: Card



My Analysis for Comparison of Total Sales Amount and Target

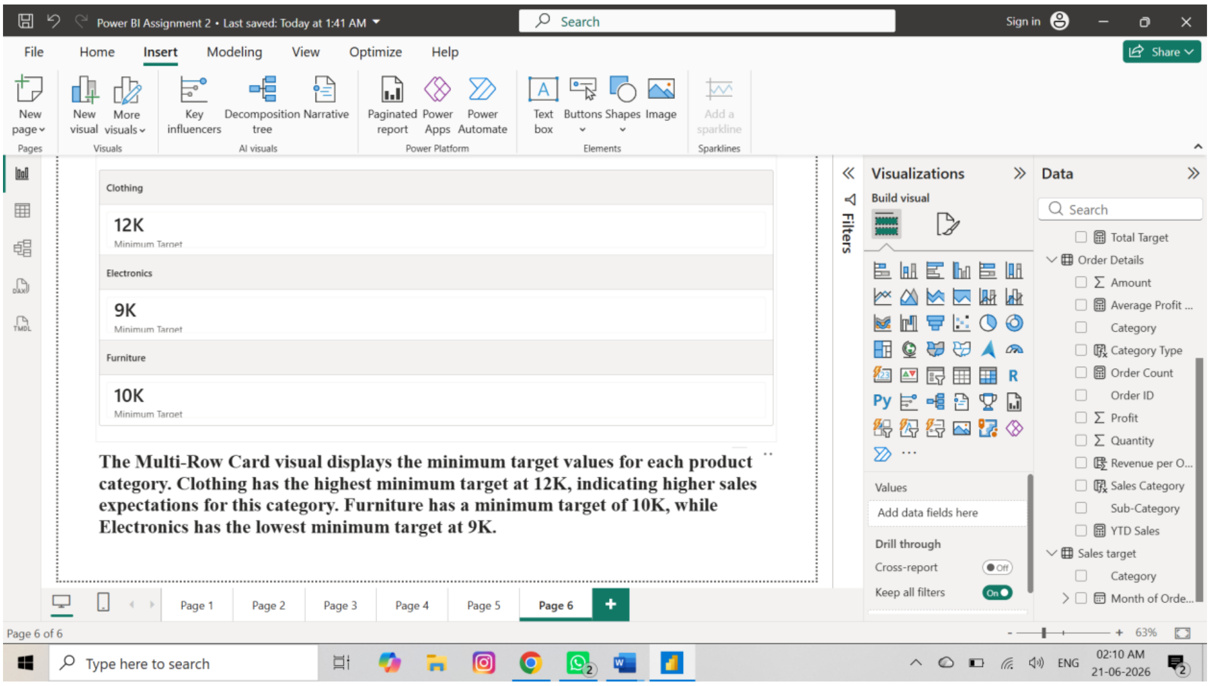
- The Total Sales achieved is 432K.
- The Total Target is 436K.
- Sales performance is very close to the target, with a difference of only 4K.
- The achievement percentage is approximately 99.1% (432K ÷ 436K × 100).
- This indicates that the business nearly met its sales goal, showing strong overall performance.
- A small improvement in sales would be sufficient to achieve or exceed the target.

Multi-Row Card – Minimum Target for Each Segment

Measure

```
Minimum Target =
MIN('Sales Target'[Target])
```

Visual Type: Multi-Row Card



My Analysis for Multi-Row Card – Minimum Target for Each Segment

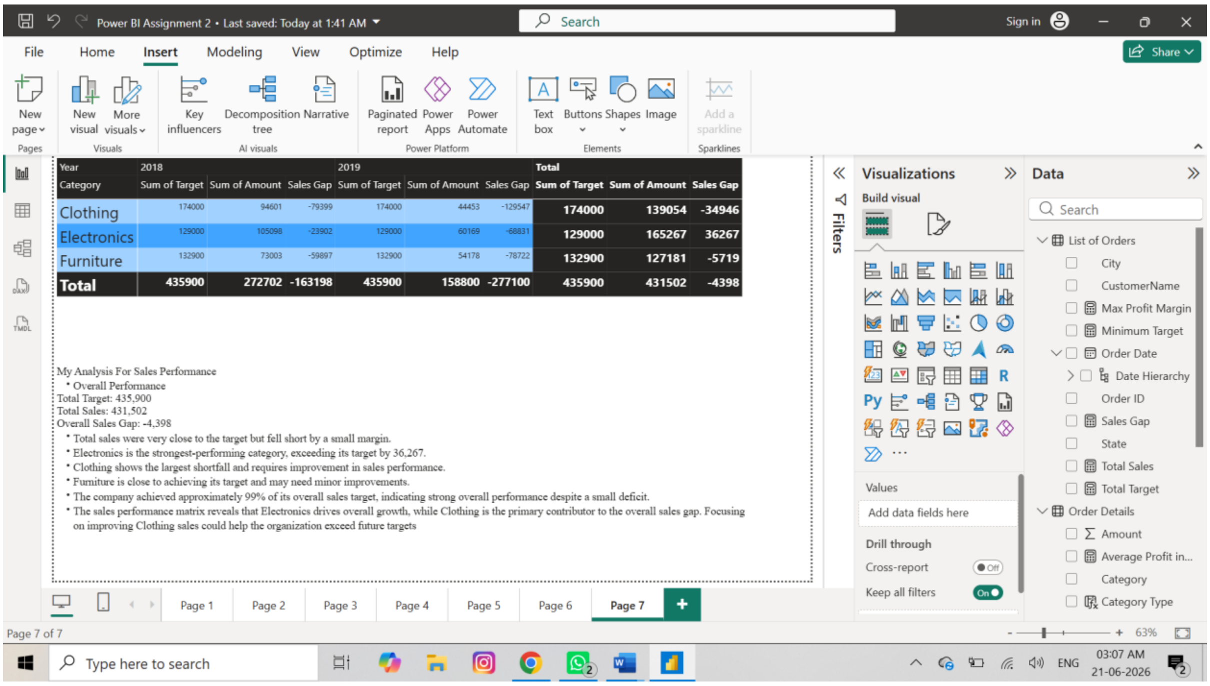
- The Multi-Row Card visual displays the minimum target values for each product category. Clothing has the highest minimum target at 12K, indicating higher sales expectations for this category. Furniture has a minimum target of 10K, while Electronics has the lowest minimum target at 9K.

Sales Performance Matrix:

Visual :Matrix

Measures

```
Sales Gap = [Total Sales]-[Total Target]
```

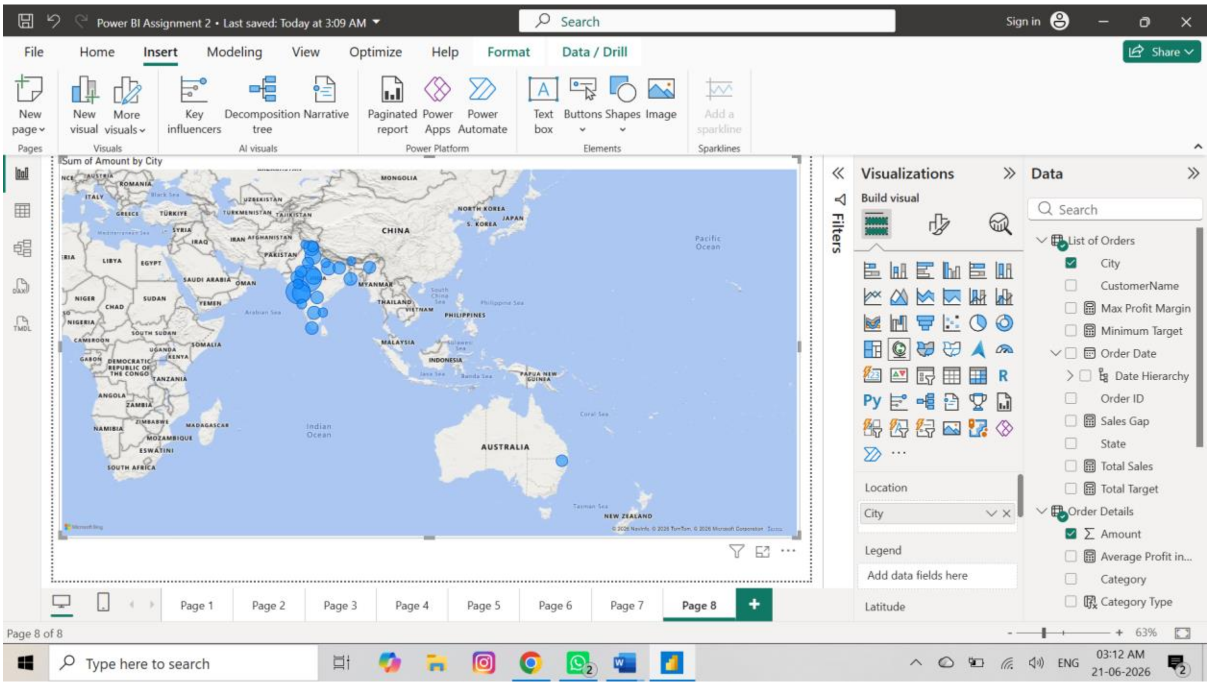


My Analysis For Sales Performance

- Overall Performance
  Total Target: 435,900
  Total Sales: 431,502
  Overall Sales Gap: -4,398
- Total sales were very close to the target but fell short by a small margin.
- Electronics is the strongest-performing category, exceeding its target by 36,267.
- Clothing shows the largest shortfall and requires improvement in sales performance.
- Furniture is close to achieving its target and may need minor improvements.
- The company achieved approximately 99% of its overall sales target, indicating strong overall performance despite a small deficit.
- The sales performance matrix reveals that Electronics drives overall growth, while Clothing is the primary contributor to the overall sales gap. Focusing on improving Clothing sales could help the organization exceed future targets

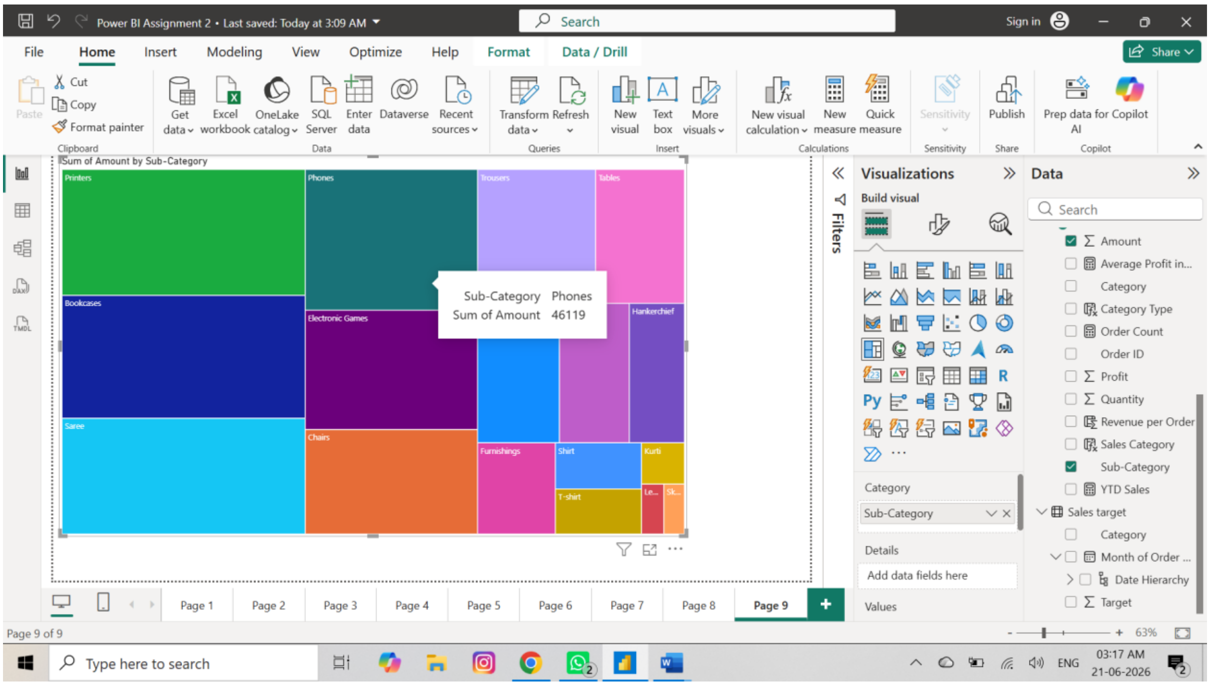
Geographic Sales Analysis

Visual: Map



Sales Distribution by Sub-Category

Visual :Treemap



My Analysis

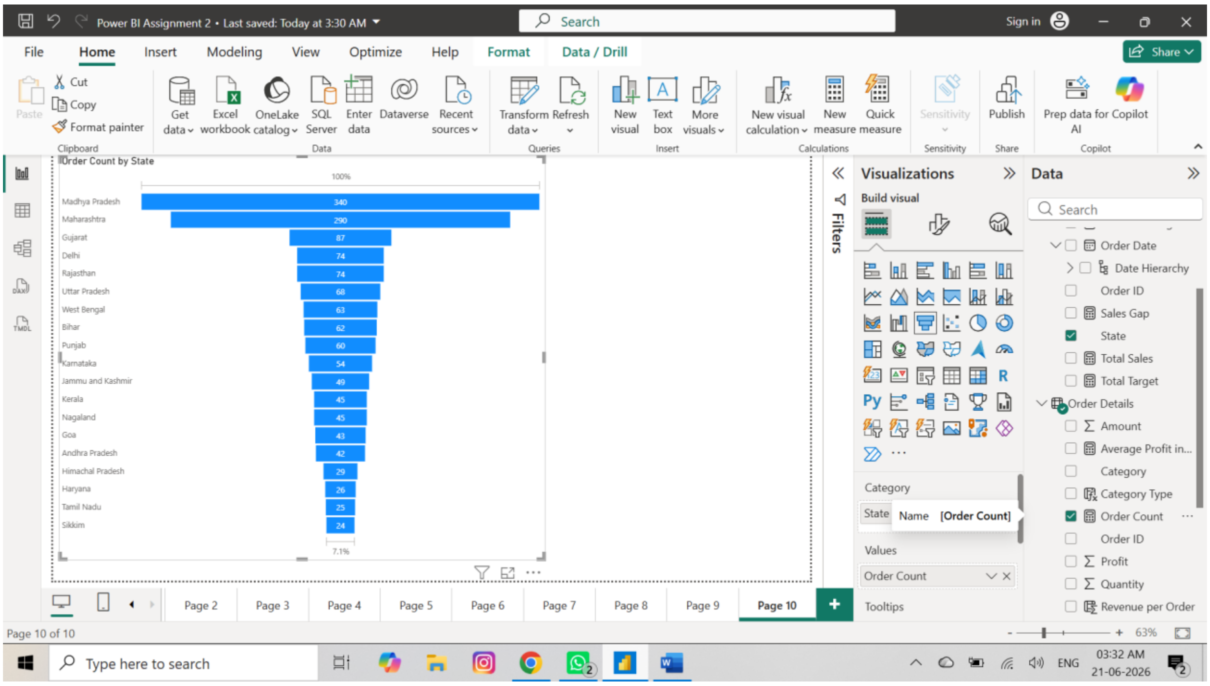
The size of each rectangle represents the sales contribution of that sub-category.

- Printers have the highest sales contribution, as shown by the largest area in the treemap.

- Bookcases, Sarees, Phones, and Electronic Games also contribute significantly to overall sales.
- Medium-performing sub-categories include Chairs, Accessories, Trousers, and Tables.
- Smaller categories such as Kurta, Leggings, and Skirts contribute less to total sales, indicating lower demand or fewer transactions.

Order Count Analysis by State

Visualization Type: Funnel Chart



Dashboard

My Analysis For Dashboard

1. Sales Target Achievement by Category

- Electronics exceeded its sales target and is the best-performing category.
- Furniture is close to its target, showing stable performance.
- Clothing did not achieve its target and requires improvement.
- Focus on Clothing sales can improve overall business performance.

2. Max Profit Margin by Sub-Category

- Bookcases, Tables, and Printers generate the highest profit margins.
- Most profit is concentrated in a few sub-categories.
- Several sub-categories contribute only a small share of profit.
- The company should focus on high-profit products to maximize earnings.

3. Monthly Sales Trend

- Sales were highest in Quarter 1.
- Sales declined from Q1 to Q3, indicating weaker demand.
- Q4 shows recovery with increased sales.
- Seasonal factors or promotions may have influenced performance.

4. Comparison of Profit and Quantity by Sub-Category

- Printers, Tables, and Bookcases produce the highest profits.
- Higher quantity sold does not always result in higher profit.
- Some products generate strong profits even at moderate sales volumes.

- Product profitability should be considered along with sales quantity.

5. Comparison of Total Sales Amount and Target

- Total Sales = 432K.
- Total Target = 436K.
- Sales achieved approximately 99.1% of the target.
- The business narrowly missed the target by only 4K, indicating strong performance.

6. Multi-Row Card – Minimum Target for Each Category

- Clothing has the highest minimum target (12K).
- Furniture has a minimum target of 10K.
- Electronics has the lowest minimum target (9K).
- This shows higher sales expectations for the Clothing category.

7. Sales Performance Matrix

- Overall Sales = 431,502 against Target = 435,900.
- Overall sales gap is only -4,398.
- Electronics exceeded its target by 36,267.
- Clothing has the largest sales shortfall.
- Furniture is close to its target.
- Overall performance is strong despite a small deficit.

8. Geographic Sales Analysis

- Sales are concentrated in major Indian cities.
- Western and Central regions contribute significant revenue.
- Some cities generate substantially higher sales than others.
- Geographic analysis helps identify strong and weak sales regions.

9. Sales Distribution by Sub-Category (Treemap)

- Printers, Bookcases, Phones, and Trousers contribute the largest share of sales.
- Smaller blocks represent lower-selling sub-categories.
- Sales are unevenly distributed across products.
- High-performing sub-categories drive overall revenue growth.

10. Order Count Analysis by State (Funnel Chart)

- Madhya Pradesh has the highest order count (340).
- Maharashtra is the second-highest with 290 orders.
- Most other states have significantly lower order volumes.
- The business has strong customer demand in a few key states.
- Marketing efforts can target low-order states to increase sales.

Conclusion:

The Power BI dashboard provides a comprehensive analysis of sales performance, targets, trends, and regional distribution. The Sales Target Achievement analysis shows that Electronics exceeded its target, while Clothing and Furniture fell short of their sales targets, resulting in an overall negative sales gap. The Sales Performance Matrix effectively compares actual sales against targets across categories and years, helping identify areas requiring improvement.

The Monthly Sales Trend reveals that sales were highest in Quarter 1, declined during Quarter 2 and Quarter 3, and recovered in Quarter 4, indicating seasonal fluctuations in demand. The Geographic Sales Analysis highlights major sales contributions from key cities and regions, enabling better understanding of regional market performance.

The Treemap and Donut Chart visualizations show that sub-categories such as Printers, Bookcases, Phones, and Sarees contribute significantly to total sales and profit margins, while several smaller sub-categories contribute less. These insights help identify top-performing products and business opportunities.